

Instacart Market Basket Analysis

Low-Level Design (LLD)

Detailed Process Flow

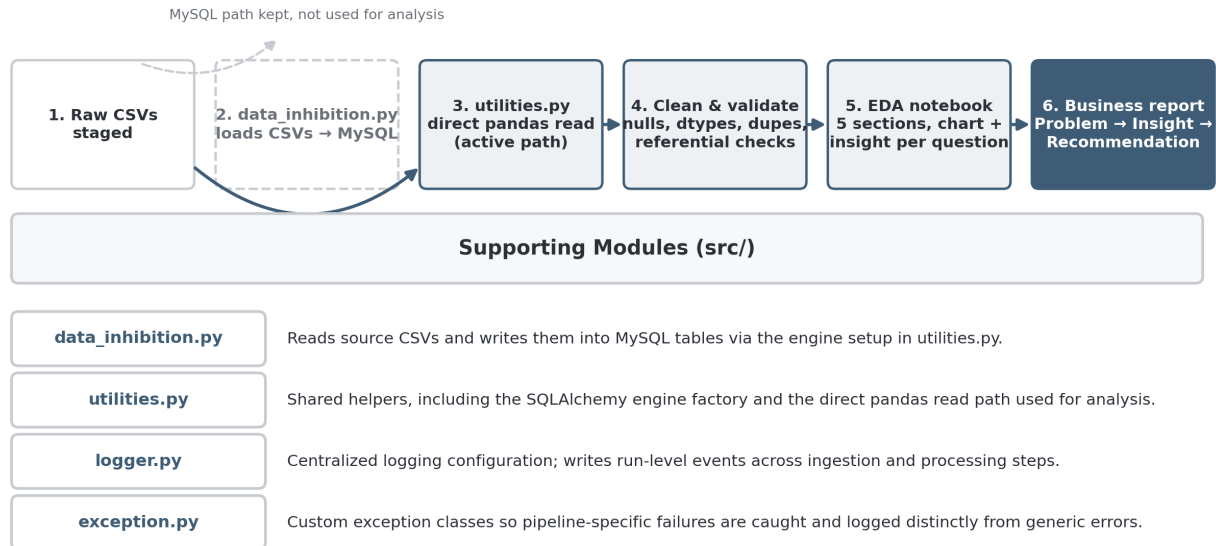


Figure 2. Step-by-step flow with module-level detail.

Data Cleaning and Validation

- Null checks across all source tables, with expected nulls documented rather than treated as defects (for example, a missing `days_since_prior_order` on a customer's first order).
- Data type validation and correction across numeric and categorical fields.
- Duplicate checks on primary identifiers (order id, product id).
- Referential checks confirming foreign keys resolve correctly across joined tables.

Exploratory Analysis Structure

The EDA notebook is organized into five thematic sections, each containing one or more business questions with a supporting chart and a written insight: Product and Category Performance, Customer Purchase Behavior, Temporal and Seasonality Patterns, Reorder Behavior, and Department and Aisle Segmentation.

Reporting

Findings are consolidated into a business-facing report structured around a problem statement, the insights that address it, and a set of recommendations for stakeholders — connecting each chart back to a concrete action rather than presenting it in isolation.